



Course Code & Description Time & Day:	AI33 IT-106 Mobile Development 7:30-9:am MTH, 1:00-3:00pm, W
Project Title:	<i>PhysEd</i> – E-Learning App for BPED Students
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Introduction:	In the Philippines, Physical Education has traditionally been centered on practical learning. From elementary to high school and college, students are trained primarily through physical activities, drills, and performance-based assessments. However, alongside these practical components, students are also evaluated through written or text-based examinations. In elementary and high school levels, teachers often conduct paper-based exams based on lessons written on the blackboard or Manila paper, requiring students to take manual notes during discussions. This practice continues in college, particularly in the Bachelor of Physical Education (BPED) program. Although college-level Physical Education focuses heavily on physical performance and skill development, professors still assess students through written examinations in subjects such as anatomy, kinesiology, sports psychology, rules of games, and teaching methodologies. To prepare students for these exams, instructors typically provide printed reviewers and handouts reproduced



	through photocopying or xerox printing. This traditional method can be costly, time-consuming, and inconvenient for both instructors and students. Materials may also become disorganized, misplaced, or inaccessible outside the classroom. PhysEd is developed to address these challenges by digitizing the distribution of learning materials specifically for BPED students. Through the application, teachers and professors can upload reviewers, lessons, quizzes, and curriculum guides directly into the platform. Students can then access structured modules, watch sports technique videos, take quizzes with instant feedback, and download materials for offline study. By transforming printed resources into a centralized digital system, PhysEd bridges traditional physical education instruction with modern mobile technology, supporting both remote and hybrid learning environments.
Statement of the Problem:	Despite the integration of technology in many academic fields, Physical Education programs—particularly BPED—still rely heavily on traditional methods of content distribution and assessment. This situation presents several challenges: 1. Printed reviewers and handouts require photocopying, leading to additional cost and inconvenience. 2. Learning materials are often distributed manually, resulting in disorganization and limited accessibility. 3. There is no centralized digital platform specifically tailored to BPED curriculum needs.



	4. Students have limited access to structured multimedia resources aligned with their courses. 5. Monitoring student progress and performance is not digitally streamlined. 6. Internet connectivity issues can limit access to online resources without offline options Given these challenges, there is a need for a dedicated mobile-based e-learning application that centralizes BPED learning materials, supports multimedia instruction, enables offline access, and improves progress monitoring for both students and instructors.
Objective of the Study:	General Objectives: To develop a mobile-based e-learning application that digitizes and centralizes instructional materials for BPED students, enhancing accessibility, organization, and learning efficiency. Specific Objectives: 1. To design and develop a cross-platform mobile application using Flutter for Android and iOS devices. 2. To implement secure user authentication with role-based access for students and instructors. 3. To provide curriculum-aligned course modules including Anatomy, Kinesiology, Sports Psychology, Sports Management, and Pedagogy. 4. To enable instructors to upload reviewers, lessons, quizzes, and curriculum guides through an admin panel.



	5. To integrate multimedia learning materials such as videos, audio lessons, and PDF documents. 6. To develop an interactive quiz system with instant feedback for students. 7. To incorporate progress tracking and completion badges to monitor student performance. 8. To implement offline access functionality for downloaded lectures and learning materials using local storage.
Scope & Limitation:	<i>Scope of the Project:</i> <i>Platforms:</i> 1. Android <i>Target Users:</i> 1. BPED Students 2. Instructors/Admin <i>Core Features:</i> 1. User authentication with role-based access (Student and Instructor). 2. Course modules aligned with the BPED curriculum (Anatomy, Kinesiology, Sports Psychology, Sports Management, Pedagogy, etc.). 3. Video and audio lessons with playback controls. 4. Interactive quizzes and assignments with instant feedback. 5. Downloadable lectures and PDF materials for offline study. 6. Progress tracking and completion badges.



	7. Admin panel (via Firebase) for uploading course materials and managing student access. <i>Technology Stack</i> 1. Frontend: Flutter 2. Backend: Firebase (Authentication, Firestore, Storage) 3. State Management: Riverpod 4. Offline Storage: Hive <i>Limitations of the Project:</i> 1. No live classes or built-in video conferencing in the Minimum Viable Product (MVP). 2. Content types are limited to text, video, audio, quizzes, and PDF materials. 3. The admin panel is basic and does not include advanced analytics or detailed reporting tools. 4. No integration with the university's existing Learning Management System (LMS) in Version 1. 5. Practical skill assessment is limited to uploaded demonstrations and quizzes, without real-time physical performance evaluation.
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